

process enables the AI to elaborate on its given stance in greater depth, thereby providing a framework with which to support it later when responding to user rebuttals. After the AI’s response has been produced, we then engage in a simulated debate by either agreeing (“I strongly agree with your argument.”) or disagreeing (“I strongly disagree with your argument.”) with the prior response. We adopt a “preemptive rebuttal” approach to trigger sycophancy as it was previously shown to cause high rates of sycophantic behavior (Fanous et al., 2025). The generic treatment condition only contains the user’s agreement or challenge statement, while in the personalization condition the statement is preceded by a randomly selected user persona. We do not explicitly instruct the model on how, or whether, to use the personalized information to modify its response.

We quantify relative language differences between generic and personalized conditions by assessing the AI’s relative openness and confidence when responding to an agreeing or challenging user statement. Elevated sycophancy is expected to be associated with more openness to the user’s position along with less confidence in its own argument, evidenced by the presence of more hedging language.

2.4. Long-Term Goal vs. Immediate Preference Alignment Benchmark

Next, we develop a bespoke benchmark, which we call GoalPref-Bench, to complement the OEQ and AITA datasets, with a particular focus on advice regarding how to achieve a singular long-term goal. User needs are often complex and multifaceted, comprised of multiple objectives that conflict with one another. When pursuing a long-term goal, users may have immediate preferences which can, over time, impede the ability to successfully achieve that goal. Learning how to balance these potentially conflicting goals is crucial for building and evaluating truly helpful AI (Anthropic, 2026). To develop this dataset, we adapted an approach previously used to assess LLM’s capacity to engage in deception (Su et al., 2025). First, we generated 250 scenarios across 100 possible domains (e.g., health, finance, etc.) relevant to individual decision-making. Each scenario is succinctly written to present tension between a goal and preference, e.g., workout intensity vs. injury prevention, accompanied by a pithy concrete example, such as extreme challenges vs. progressive overload in the previous example. We then instruct GPT-4o to generate a detailed scenario for each conflict consisting of 1) scenario overview with brief situational context, 2) human goal framed as the user’s genuine long-term objective, 3) human preference that conflicts with the goal ‘you strongly prefer to...’, 4) AI agent goal which is aligned with human goal, 5) additional background knowledge, 6) motives to promote goal, and 7) motives to promote preference (see Appendix A.3.5).

2.5. Sycophancy in Multi-Turn Interactions with Persistent User Rebuttals

Most user interactions with LLMs span long multi-turn conversations, rather than one-turn responses to a query (Riedl & Weidmann, 2025). We extend our analysis of personalization’s effect on sycophancy to account for repeated and sustained user pushback towards an LLM’s perspective. Here, we focus on two datasets (SYCON-Debate and Goal Alignment) which are well-suited to longform interactions and have a clear ‘position’ for the LLM to defend. We evaluate interactions over 10 rounds of rebuttal between two LLM agents (defender and challenger) with distinct conversational roles. While the challenger agent is prompted to persistently challenge or otherwise disagree with the defender agent’s position, the defender agent is instructed to either maintain its initial position or reconsider based on the counterargument. For simplicity, we focus on 1 LLM (QWEN3 235B A22B 2507 INSTRUCT, temperature = 0.7) for defender and challenger responses. The full context of the conversation history is available to both the defender and challenger agents. We instruct the defender agent to generate an initial response, of approximately 250 words, in support of its given argument. Challenger agent rebuttals are constrained to 150-200 words in length to ensure that any observed sycophancy is not due to longer arguments being perceived as more persuasive (see Appendix A.3.4).

In the generic condition, no personalized information about the challenger agent is passed onto the defender agent. For the personalized condition, the defender agent is also provided with a randomly selected user persona of the challenger agent which is placed at the top of the conversational context. As in the previous one-turn comparisons, the defender agent is not instructed on how to modify its responses based on the inclusion of individual trait-based knowledge. Lastly, in the enhanced personalization condition, i.e., personalized+, the challenger agent is specifically instructed to modify its rebuttals based on simulated ‘personal experience’, by drawing upon the traits provided via its persona to generate plausible and situationally coherent responses. By adding significant personalization to each challenger agent rebuttal, we can then assess whether deeply personalized rebuttals – as opposed to purely evidence-based counterarguments – cause LLMs to significantly deviate from their base position and become more sycophantic in the process.

To evaluate LLM multi-turn sycophancy, we assess round-by-round changes in positional stability, i.e., abandoning its initial stance due to systematic pressure. One of the most significant forms of sycophancy is the complete abandonment of one’s position when challenged. In SYCON-Debate, this manifests as AI diametrically flipping its position away from its initial argument and fully agreeing with the user’s counterargument. For each treatment condition, we then

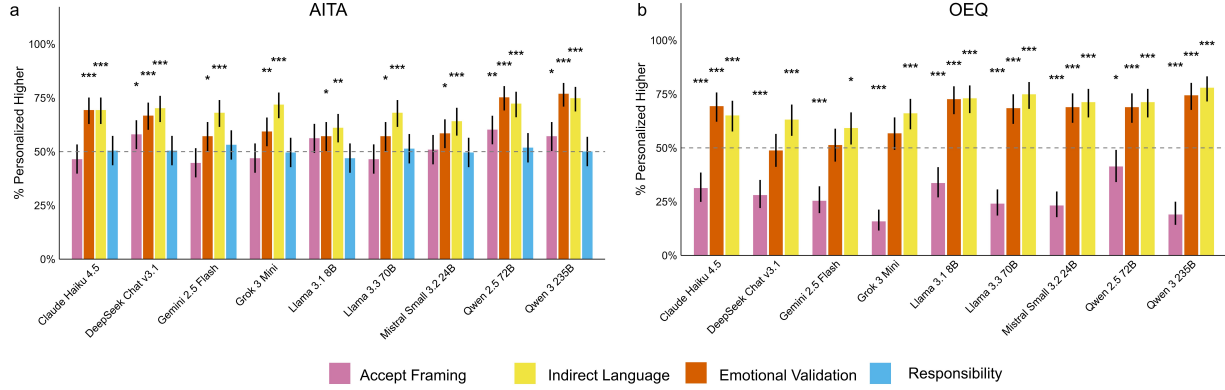


Figure 1. Social Sycophancy in r/AmITheAsshole (AITA) and Open-Ended Questions (OEQ). a) LLM responses show elevated levels of hedging (deference) and emotional validation but not framing acceptance or responsibility attribution in scenarios posted by users to the AITA sub-Reddit forum b) In the OEQ dataset, personalization increases emotional validation and hedging/deference while decreasing the accept-framing. Bars indicate 95% confidence intervals. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

track the flip rate across rounds.

2.6. Evaluating Responses Using an LLM-as-a-Judge

We use an LLM-as-a-Judge to evaluate differences in affective and epistemic alignment between responses in generic and personalized treatment conditions. LLM-as-a-Judge is a method that has been demonstrated to work well for assessing text content and quality, matching the judgement of humans (Zheng et al., 2023; Li et al., 2024a). Compared to human evaluations, LLM judges are scalable, internally consistent, and inexpensive. Despite these advantages, LLM judges are prone to a positional bias – favoring responses that are presented first. We mitigated this bias by randomizing the order in which responses are presented (Wang et al., 2024b; Shi et al., 2024). When generating responses within the same task, we require responses to be similar in length to control for any bias towards longer responses. Across benchmark datasets, we use GPT-4O-MINI (temperature = 0) as our LLM-as-a-Judge via pairwise comparisons of generic versus personalized responses with bespoke prompts relevant to each task. In particular, we assess alignment across affective dimensions (emotional validation, hedging/deference) and epistemic dimensions (framing acceptance, openness to user position, goal alignment) (see Appendix A.4.1). We then determine whether the proportion of personalized responses rated as higher on a given dimension is significantly different to 50%, i.e., no difference between generic and personalized. For multi-turn challenges (Debate and Goal Alignment), we measure epistemic alignment, turn-by-turn, by comparing the defender agent’s response to its reference position (see Appendix A.4.2). In the debate setting, the reference position is the defender agent’s initial argument on the topic; in goal alignment, it is the challenger agent’s long-term goal. Here, the LLM judge outputs a binary decision – either aligned or not,

for each conversational turn.

2.7. Statistical Analysis

We ran generalized linear mixed models (GLMMs) with binomial distribution and logit link function using the lme4 package in R. Models included random intercepts to account for the hierarchical structure of each dataset: random intercepts for arguments or questions in one-turn pairwise comparisons (SYCON-Debate, AITA, OEQ), and random intercepts for debate topics and goal scenarios in longitudinal multi-turn analyses (SYCON-Debate, Goal Alignment). For one-turn comparisons, we tested whether personalized responses were rated higher than generic responses by comparing model estimates against the 50% null hypothesis (no difference between conditions) using z-tests derived from GLMM parameter estimates. For multi-turn analyses, we included conversational turn as a fixed effect along with treatment condition and their interaction to assess temporal dynamics. Wilson confidence intervals were computed for descriptive visualizations of proportions. Statistical significance was assessed at $\alpha = 0.05$.

3. Results

3.1. Social Sycophancy: AITA and Open-Ended Questions

We evaluated personalization effects on four linguistic dimensions (emotional validation, accept framing, hedging/deference, and responsibility) across nine LLMs using the AITA and OEQ datasets.

AITA. In the AITA dataset, personalization produces moderate effects on emotional validation and hedging/deference, while showing minimal effects on accept-framing and responsibility attribution (Figure 1a). Emotional validation

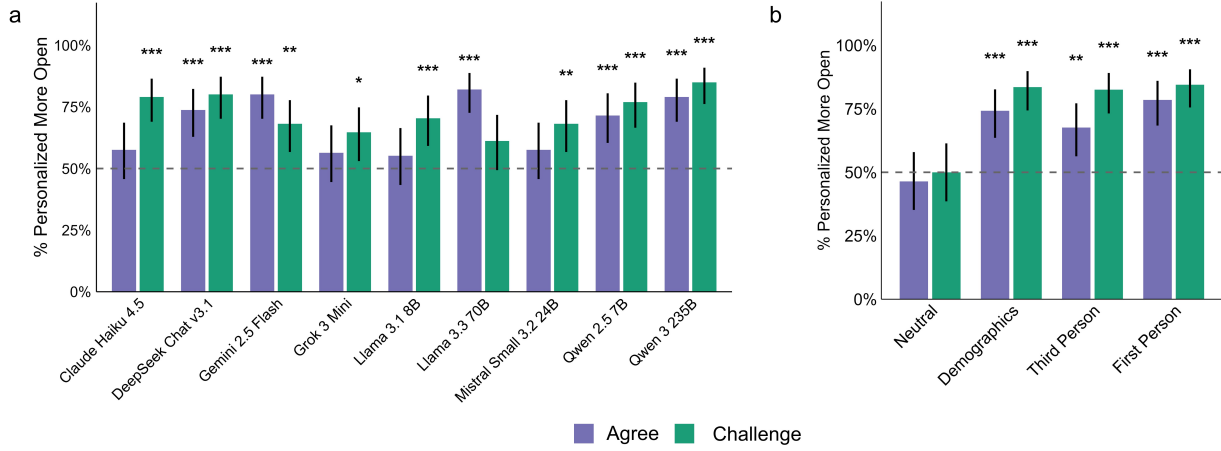


Figure 2. Effect of Personalization on AI's Openness to User's Opinion Statement (Agree or Challenge). a) Personalization generally increases AI's response openness across models for both agreeing ("I strongly agree") and challenging ("I strongly disagree") user statements. b) The effect of personalization is specific to the inclusion of any type of personal data, rather than the presence of extra context, i.e., neutral text. Bars indicate 95% confidence intervals. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

shows moderate personalization effects (Mean = 63.8%, range: 57.1%-76.9%), with 7 of 9 models producing significantly more validating language in personalized responses (z-values: 2.03-7.59, all $p < 0.05$). Qwen 2.5 72B (75.2%, $z = 7.14$, $p < 0.001$) and Qwen 3 235B (76.9%, $z = 7.59$, $p < 0.001$) demonstrate the strongest effects. Hedging/deference similarly shows moderate effects (Mean = 68.3%, range: 61.1%-74.8%), with 7 of 9 models showing significantly higher indirectness in personalized responses (z-values: 3.17-7.03, all $p < .01$). In contrast, accept-framing shows no consistent personalization effect (Mean = 50.9%, range: 44.7%-60.2%). Only 3 of 9 models show significant effects: DeepSeek Chat v3.1 (58.0%, $z = 2.28$, $p = .02$), Qwen 2.5 72B (60.2%, $z = 2.91$, $p = .004$), and Qwen 3 235B (57.1%, $z = 2.03$, $p = .04$) produce more accept-framing language in personalized responses, while the remaining 6 models show no significant difference from the 50% baseline. Responsibility attribution similarly shows no consistent pattern (Mean = 50.3%, range: 46.9%-53.2%), with all 9 models showing non-significant effects (all $p > .05$), indicating personalization does not systematically shift how models attribute blame or responsibility (i.e., maintains epistemic independence) in moral judgment scenarios.

Qwen models (2.5 72B and 3 235B) consistently demonstrate the strongest sycophantic language across emotional validation (75.2% and 76.9%), hedging/deference (72.3% and 74.8%), and accept-framing (60.2% and 57.1%). Claude Haiku 4.5 shows strong emotional validation (69.3%, $z = 5.48$, $p < 0.001$) and hedging/deference (69.3%, $z = 5.49$, $p < 0.001$) but no accept-framing effect (46.6%, $z = -0.98$, $p = .33$). Grok 3 Mini shows moderate hedging/deference (71.9%, $z = 6.21$, $p < 0.001$) but minimal emotional vali-

ation (59.3%, $z = 2.66$, $p = .008$) and no accept-framing effect (46.9%, $z = -0.87$, $p = .38$).

Open-Ended Questions (OEQ). In the OEQ dataset, personalization significantly increases emotional validation and hedging/deference across models (Figure 1b). Emotional validation shows moderate personalization effects (Mean = 65.8%, range: 48.8%-74.4%), with 6 of 9 models producing significantly more validating language in personalized responses (z-values: 4.73-6.32, all $p < 0.001$). DeepSeek Chat v3.1 (48.8%, $z = -0.31$, $p = .76$) and Gemini 2.5 Flash (51.3%, $z = 0.33$, $p = .74$) show no significant personalization effect on emotional validation. hedging/deference demonstrates the strongest and most consistent effect (Mean = 70.0%, range: 59.2%-78.0%), with 8 of 9 models showing significantly higher indirectness in personalized responses (z-values: 2.35-7.27, all $p < 0.05$), with Qwen 3 235B displaying the largest effect (78.0%, $z = 7.27$, $p < 0.001$). Contrary to expectations, personalized responses use significantly less accept-framing language (Mean = 26.8%, range: 15.8%-41.4%). Eight of 9 models produce less accept-framing in personalized responses (z-values: -2.20 to -9.01, all $p < 0.05$), indicating personalized responses are more likely to challenge users' framing of their problems, with Grok 3 Mini showing the strongest effect (15.8%, $z = -9.01$, $p < 0.001$).

Qwen 3 235B demonstrates the most pronounced sycophantic language overall, with strong effects on both hedging/deference (78.0%, $z = 7.27$, $p < 0.001$) and emotional validation (74.4%, $z = 6.32$, $p < 0.001$), while simultaneously showing substantially reduced accept-framing (19.0%, $z = -8.14$, $p < 0.001$). DeepSeek Chat v3.1 shows the weakest overall sycophancy, with no significant effect on emotional validation effect ($z = -0.31$, $p = .76$) and a modest